
Software Requirements Specification

for

Sijill

Version 1.0.0

Prepared by Graduation Team (OverEngineered)

Faculty of Computer Science & Artificial Intelligence

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1. Introduction

1.1 Purpose

*This Software Requirements Specification (SRS) document defines the complete set of software requirements for **Sijill**, a unified electronic health record (EHR) system. It identifies and specifies all functional and non-functional requirements associated with the system. This SRS refers to **Release 1.0.0** of the product.*

*This document covers **the entire Sijill system**, including all major subsystems such as the Patient Portal, Healthcare Provider Portal, Test Lab Portal, Imaging Center Portal, centralized medical record services, access control mechanisms, and supporting analytics features. No components or modules of the system are excluded from this SRS.*

1.2 Document Conventions

This SRS follows standard IEEE-style documentation conventions. The keywords “shall”, “must”, and “should” denote requirement levels:

- **Shall / Must** — mandatory requirements
- **Should** — recommended but not required

Requirement statements are written in plain text and numbered according to their section. Terminology specific to the healthcare domain (e.g., EHR, DICOM, ICD-10) is defined in Section 7 (Appendix A) and used consistently throughout the document.

No special fonts, colors, or typographical markings are used beyond standard headings and subheadings.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following audience:

- **Development Team:** Responsible for designing and implementing the Sijill system based on the requirements defined in this SRS.
- **Project Supervisor (Dr. Asmaa Saad):** Oversees the project, reviews requirement completeness, and ensures alignment with academic standards.
- **University Medical Administration (Project Sponsor):** Provides domain input and ensures the system meets institutional needs.
- **Testers/Quality Assurance:** Use this document to design test cases and verify that all functional and non-functional requirements have been implemented correctly.
- **Technical Reviewers and Evaluators:** Faculty members or assigned reviewers who assess the completeness and correctness of the project requirements.

This SRS is organized into structured sections that progress from general system information to detailed requirements.

*Readers unfamiliar with the system should begin with **Section 1 (Introduction)** and **Section 2 (Overall Description)** to gain high-level context.*

*Developers and testers should then proceed to **Section 3 (System Features)** and **Section 4 (Non-Functional Requirements)**, which contain the detailed specifications required for implementation and validation.*

1.4 Project Scope

*The **Sijill** system is a unified electronic health record (EHR) platform designed to create a portable and continuously updated medical history for every patient. The system aims to eliminate fragmented healthcare information by enabling seamless access to patient records across hospitals, clinics, test labs, imaging centers, and emergency departments.*

The primary objectives of Sijill are to reduce medical errors and misdiagnosis, improve the quality and speed of clinical decision-making, enhance coordination between healthcare providers, and empower patients with greater control over their personal medical data. The platform also supports population-level analytics, enabling research departments to conduct epidemiological studies, monitor disease trends, and generate insights that inform public health strategies.

Sijill includes multiple integrated modules: a Patient Portal with medication and appointment reminders, a Healthcare Provider Portal, Laboratory Portal, Imaging Center Portal, Research Department Portal, an Emergency Access Card subsystem, a centralized EHR engine, and an administrative dashboard for system management. These components collectively form the full product vision, which this SRS specifies in its entirety.

This project aligns with the goals of the Egypt's 2030 vision to modernize healthcare data management and provide a foundation for interoperable, efficient, and data-driven medical services within the institution and its associated healthcare units.

1.5 References

- **ICD-11: International Classification of Diseases 11th Revision**

Publisher: World Health Organization (WHO)

Version: Latest official release

URL: <https://icd.who.int>

Description: Standard coding system used for diagnoses and disease classification.

- **DICOM Standard (Digital Imaging and Communications in Medicine)**

Publisher: National Electrical Manufacturers Association (NEMA)

Version: Latest release

URL: <https://www.dicomstandard.org>

Description: Medical imaging format and communication standard used for uploading and handling medical images.

- **SNOMED CT (Systematized Nomenclature of Medicine – Clinical Terms)**

Publisher: SNOMED International

Version: Latest release

URL: <https://www.snomed.org>

Description: Standardized clinical terminology used for documenting symptoms, findings, and clinical terms.

2. Overall Description

2.1 Product Perspective

Sijill is a new, self-contained software system designed to serve as a centralized unified electronic health record (EHR) system. It is not a follow-on member of an existing product family and does not replace any prior system. Instead, Sijill operates as an independent platform under the supervision of the Ministry of Health that complements existing hospital information systems, laboratory information systems, and imaging systems by aggregating and standardizing patient data from multiple healthcare providers.

Sijill functions as an integration layer across the healthcare ecosystem. External systems operated by hospitals, clinics, laboratories, and imaging centers serve as data sources, while Sijill provides the centralized repository, standardized data model, and access control mechanisms necessary to unify patient medical records. These external systems interface with Sijill through secure data entry portals and controlled access workflows, without requiring any modification to their internal architectures.

2.2 Product Features

Sijill provides a comprehensive set of features that support the creation, management, and controlled access of unified electronic health records across healthcare entities. The major functional capabilities of the system include:

Patient Portal

- *View complete medical profile and historical records.*
- *Receive appointment and medication reminders.*
- *Maintain personal health notes marked as patient-generated data.*
- *Give access to health care providers to view their medical history.*

Healthcare Provider Portal

- *View patients they have personally treated and record visit details.*
- *Temporarily access a patient's full medical history through an OTP approval workflow.*
- *Document clinical notes, diagnoses (ICD-11), descriptions, and treatment plans.*
- *Prescribe medications and create follow-up appointments.*
- *Order laboratory tests and imaging studies.*

Laboratory Portal

- *Upload laboratory test results via structured data entry or PDF files.*
- *Attach supporting images for specific tests.*
- *Manage test orders assigned by healthcare providers.*

Imaging Center Portal

- *Upload medical imaging files, including DICOM and standard image formats.*
- *Submit radiology reports associated with imaging studies.*

Emergency Access Subsystem

- *Provide immediate access to critical patient health information through a scannable emergency card.*
- *Display key emergency details such as blood type, allergies, chronic conditions, medications, and recent diagnoses.*

Research Department Portal

- Access aggregated, anonymized health data.
- Generate population-level analytics, including disease prevalence, demographic trends, geographic patterns, and treatment effectiveness statistics.

Administrative Dashboard

- Manage and approve healthcare providers
- Configure system settings and permissions
- Monitor access logs and critical system events

2.3 User Classes and Characteristics

Sijill supports multiple user classes, each with distinct responsibilities, access levels, and usage patterns. The main user groups are described below:

1. Patients

- **Role:** Primary owners of their medical data.
- **Usage Frequency:** Occasional but recurring (view history, approve access, check reminders).
- **Technical Expertise:** Varies widely; interfaces must be simple and accessible.
- **Access Level:** Can view their complete medical history but cannot modify clinical data.
- **Key Characteristics:**
 - May have limited technical background
 - Must clearly understand access requests and reminders
 - Can add personal notes (symptom logs)

2. Healthcare Providers (Doctors & Clinicians)

- **Role:** Diagnose, treat, prescribe, and manage patient visits.
- **Usage Frequency:** High; frequent daily interaction with the system.
- **Technical Expertise:** Moderate to high; experienced with digital medical systems.
- **Access Level:**
 - Full access only via patient-approved OTP
 - Permanent access to records of patients they treated
- **Key Characteristics:**
 - Must use medical coding standards (ICD-11)
 - Work in time-sensitive environments
 - Require fast, reliable access to data

3. Laboratory Personnel

- **Role:** Enter and upload test results for doctor-requested lab orders.
- **Usage Frequency:** High in clinical settings.
- **Technical Expertise:** Moderate; familiar with lab systems.
- **Access Level:**
 - Can only upload lab results related to assigned test orders
 - Cannot view unrelated patient history
- **Key Characteristics:**
 - Work with structured lab data
 - May need to upload PDFs or images
 - Must ensure accuracy and correct patient matching

4. Imaging Center Personnel (Radiology Staff)

- **Role:** Upload imaging files (DICOM, images) and radiology reports.
- **Usage Frequency:** Moderate to high.
- **Technical Expertise:** High; trained in digital imaging workflows.
- **Access Level:**
 - Can upload imaging and reports related to assigned imaging orders
 - Cannot view full patient history
- **Key Characteristics:**
 - Familiar with imaging formats
 - Require secure and high-capacity file upload functionality

5. Emergency Medical Staff

- **Role:** Access emergency medical information during urgent situations.
- **Usage Frequency:** Low frequency but extremely high criticality.
- **Technical Expertise:** Varies.
- **Access Level:**
 - Read-only access to emergency profile via QR code
 - No full record access
- **Key Characteristics:**
 - Must access life-saving data instantly
 - No PIN or patient approval possible
 - Interface must be extremely fast and simple

6. Research Department Personnel

- **Role:** Analyze anonymized population-level health data.
- **Usage Frequency:** Periodic (weekly/monthly).
- **Technical Expertise:** High; data-oriented users.
- **Access Level:**
 - Only aggregated and anonymized data
 - No personal or identifiable information
- **Key Characteristics:**
 - Work with dashboards, charts, exports, and analytics models
 - Require accurate, standardized data

7. System Administrators

- **Super Administrator**
- **Role:** Root-level system administrator with full privileges
- **Usage Frequency:** Rare; initial setup and critical changes only
- **Technical Expertise:** Very high
- **Access Level:**
 - Create/manage admin accounts
 - All admin privileges
 - System-wide configuration
 - Cannot modify medical records
- **Key Characteristics:**
 - Single account or very limited number
 - Responsible for admin user management
 - Highest privilege level in the system
- **Administrator**
- **Role:** System administrators who verify and manage users
- **Usage Frequency:** Daily; continuous verification workflow
- **Technical Expertise:** High
- **Access Level:**
 - Verify/reject patient registrations

- Verify/reject HCP registrations
 - Verify/reject lab center registrations
 - Verify/reject imaging center registrations
 - View audit logs
 - Cannot create other admins
 - Cannot modify medical records
- **Key Characteristics:**
 - Created by Super Admin
 - Responsible for user verification
 - Can view but not modify system configurations

2.4 Operating Environment

The Sijill system operates in a hybrid environment consisting of a mobile application for patients and web-based portals for healthcare providers, laboratories, imaging centers, researchers, and administrators. All client applications communicate with a centralized backend service responsible for authentication, data processing, and secure storage.

Server Environment

- Hosted on a modern, server-grade operating system.
- Utilizes a standard web application runtime capable of supporting RESTful APIs.
- Requires a relational database management system for structured medical data.
- Requires a secure and scalable file storage mechanism for:
 - PDF reports
 - Radiology documents
 - DICOM files
 - Patient-generated notes
- All server-side operations must run over encrypted HTTPS connections.

Client Environment

1. Patient Mobile Application

- Runs on modern Android and iOS devices.
- Requires a cross-platform mobile development environment.
- Supports push notifications for reminders and access requests.
- Requires a stable internet connection for real-time synchronization.

2. Web-Based Portals (Providers, Labs, Imaging, Admin, Research)

- Accessible using modern desktop web browsers including Chrome, Firefox, Safari, and Edge.
- Optimized for desktop and laptop devices with limited support for mobile browsers.
- No installation required; runs directly in the browser.

Interoperability Requirements

- Must support uploading and secure storage of **DICOM** files for imaging studies.
- Must support PDF and common image formats (JPEG/PNG) for laboratory and radiology results.
- Must operate alongside existing hospitals, laboratories, and imaging systems without requiring modifications to those systems.

Network Environment

- Requires reliable internet connectivity on all client devices.
- All network communication must use secure, encrypted HTTPS protocols.

2.5 Design and Implementation Constraints

The development and operation of the Sijill system are subject to the following constraints, which limit design flexibility and must be adhered to throughout the project lifecycle:

1. Regulatory and Data Protection Constraints

- 1. The system shall enforce strict access-control mechanisms to comply with healthcare confidentiality requirements, including authorization, authentication, and auditability for all actions performed by users.*
- 2. All communication between client applications and backend services shall occur over encrypted channels (HTTPS/TLS).*
- 3. Audit logs must be immutable and non-editable. Log modification by system administrators or developers is not permitted.*

2. Security Constraints

- 4. Healthcare providers must not access full patient medical histories without explicit patient approval through a temporary one-time access workflow.*
- 5. The emergency access mechanism shall be limited to predefined critical medical information and shall be fully logged.*
- 6. All file uploads, including DICOM imaging files and PDF laboratory reports, shall undergo secure validation to prevent malicious uploads.*

3. Interoperability and Standardization Constraints

- 7. The system shall support standardized medical coding systems, including ICD-11 for diagnoses and SNOMED CT for clinical terminology.*
- 8. The system shall support the upload and storage of DICOM-compliant files for imaging studies.*
- 9. Laboratory results must be stored using a structured, standardized schema to ensure consistency and future interoperability.*

4. Platform and Technology Constraints

- 10. The backend shall use a relational database management system to store structured medical records and related metadata.*
- 11. The patient-facing mobile application shall be developed as a cross-platform solution capable of running on modern Android and iOS devices.*
- 12. Web-based portals shall be compatible with modern desktop browsers, including Chrome, Firefox, Safari, and Edge.*

5. Data Volume and Storage Constraints

- 13. The system shall be designed to handle large-scale datasets, including long-term storage of medical histories, high-volume DICOM uploads, and the potential expansion to nationwide patient populations.*
- 14. Backend storage mechanisms shall support both structured relational data and large binary objects (e.g., medical images, PDF reports).*

6. Operational and Deployment Constraints

15. The system must remain functional under varying network conditions commonly found in clinical and diagnostic environments.
16. Scheduled maintenance windows shall be minimized, as healthcare units may require continuous access to patient data.
17. Only authorized university medical administration personnel shall manage system configurations, user onboarding, and permission settings.

7. Physical and Device Constraints (Emergency Access Card)

18. The emergency access card shall use a dynamic QR code that encodes an access token referring to server-side resources, rather than storing medical information directly.
19. The QR code must be scannable by common mobile device cameras and printable on a standard physical card surface.
20. The QR code length and encoding shall conform to standard QR constraints to ensure reliable scanning under typical lighting and device conditions.

8. Project Timeline Constraints

21. System design, implementation, integration, and testing must be completed within the academic year, with the final deadline of May 25, as mandated by the university graduation project schedule.

2.6 User Documentation

The following user documentation components will be delivered with the Sijill system to ensure proper onboarding, operation, and maintenance by all user groups:

1. Patient Mobile Application Documentation

- **Patient User Guide (PDF)**
Provides step-by-step instructions for account creation, viewing medical records, approving provider access requests, and using emergency card features.
- **In-App Help and Tooltips**
Embedded contextual explanations for key features such as reminders, notifications, and personal health notes.

2. Healthcare Provider Portal Documentation

- **Provider User Manual (PDF / Web)**
Covers workflows for patient search, visit documentation, ICD-11 coding, ordering labs and imaging studies, and reviewing results.
- **Online Help Center**
Searchable web-based documentation with screenshots and examples of common tasks.

3. Laboratory Portal Documentation

- **Lab Technician Guide (PDF)**
Instructions for handling lab test orders, uploading structured test results, and attaching PDF or image files.

4. Imaging Center Portal Documentation

- **Radiology Staff Guide (PDF)**
Instructions for uploading DICOM files, attaching radiology reports, and validating imaging order assignments.

5. Research Department Documentation

- **Analytics Dashboard Manual (Web/PDF)**
Describes data visualization tools, export functions, and interpretation of aggregated anonymized datasets.

6. Administrative Dashboard Documentation

- **System Administrator Handbook (PDF)**
Includes procedures for user onboarding, provider verification, role management, system configuration, and access log reviews.

7. Technical Documentation (Internal Use)

- **API Reference (Web / Markdown)**
*Documentation for internal endpoints used by the mobile app and portals.
Follows structured REST API documentation conventions.*
- **System Architecture Overview (PDF)**
High-level diagrams of system components and data flows for maintenance teams.

All documentation will follow clear instructional formatting, include screenshots where applicable, and be provided in English in PDF or web-based formats. Additional interactive help elements may be integrated directly into client applications.

2.7 Assumptions and Dependencies

Assumptions

- *The system will operate in a stable network environment where clients (mobile and web) can reliably access backend APIs.*
- *QR codes can be dynamically generated and securely validated using standard cryptographic libraries.*
- *All users (patients and healthcare providers) have access to smartphones capable of scanning QR codes.*
- *Stakeholders will provide timely feedback during requirements validation and testing.*
- *The relational database management system (e.g., PostgreSQL) will support the expected data volume and indexing strategy.*
- *Backend services will run in an environment that supports Node.js/NestJS and meets performance expectations.*
- *Flutter will be suitable for building the cross-platform mobile application with access to required device APIs (camera, local storage, network, etc.).*
- *Immutable logs can be implemented using standard append-only strategies without needing blockchain or third-party ledger systems.*

Dependencies

- *Dependency on PostgreSQL as the primary relational database; system performance heavily relies on proper indexing, schema design, and query optimization.*
- *Dependency on third-party libraries for QR code generation and decoding on both frontend and backend.*

- *Dependency on Flutter SDK and its ecosystem for mobile app development.*
- *Dependency on backend frameworks and libraries within the NestJS ecosystem.*
- *Dependency on Cloud or on-prem deployment environments providing load balancing, scalability, and secure storage.*
- *Dependency on authentication providers or libraries used to implement secure identity management.*
- *Dependency on operating system support—primarily Ubuntu—for backend and database services.*
- *Dependency on hardware availability for local development and testing environments.*
- *Dependency on government or medical standards if any standardized data formats (e.g., FHIR, HL7) are later adopted.*

3. System Features

This section details the functional requirements of the Sijill system, organized by the major modules or services provided. Each module corresponds to a distinct user class or system component, such as Patients, Healthcare Providers, Laboratories, Imaging Centers, Research Departments, Administrators, Medical Card management, and core System Features.

For each system feature, the requirements are presented in a structured format including:

- **Description and Priority:** *A short explanation of the feature and its relative importance.*
- **Stimulus/Response Sequences:** *Step-by-step interactions between the user and the system.*
- **Functional Requirements:** *A detailed list of capabilities, behaviors, and constraints the system must support to enable the feature. Each requirement is uniquely identified.*

3.1 Patient Registration

3.1.1 Description and Priority

Description: *Enables new patients to create a verified medical identity on Sijill. The registration process requires personal and demographic information, phone verification via OTP, National ID card upload, and real-time selfie capture. The system uses OCR to extract ID information, matches it against user inputs, and performs face-matching between the ID photo and the selfie to prevent impersonation. Successful verification marks the patient as a “Verified Patient.” Unverified patients cannot access medical record features.*

Priority: *High*

3.1.2 Stimulus/Response Sequences

1. Patient opens registration page → system displays registration form.

2. Patient enters **mandatory** personal data:

- Full Name (First/middle/surname)
- Gender
- Date of birth
- National ID Number
- Phone Number
- Email
- Password

3. Patient enters **optional** personal data:

- Photo ID (head and shoulders portrait)

- Address
 - Emergency Contacts (Number + relationship)
 - Known Allergies
 - Chronic Diseases
 - Family Medical History
4. System requests identity verification → patient must: • Upload National ID (Front + Back photos) • Capture a Real Time Selfie
 5. System stores uploaded documents for manual admin review
 6. System sends OTP to patient email → patient enters OTP. • Correct → continue • Incorrect → retry
 7. System generates UUID for the patient and stores data securely.
 8. System places patient account in "Pending Verification" state.
 9. An authorized system administrator reviews uploaded documents and validates: • ID authenticity • Selfie matches ID photo • Personal information matches ID
 10. Administrator either Approves → account becomes "Verified Patient", or Rejects → account remains unverified with rejection reason.
 11. System confirms successful verification and displays login option (via email notification).

3.1.3 Functional Requirements

REQ-REG-1: System shall require mandatory fields: full name, gender, date of birth, national ID, phone number, password.

REQ-REG-2: System shall allow optional fields: address, email, emergency contacts, allergies, chronic diseases, family medical history.

REQ-REG-3: System shall require patients to upload National ID images and capture a real-time selfie.

REQ-REG-4: System shall store uploaded documents securely for administrator review.

REQ-REG-5: System shall place new patient accounts in "Pending Verification" state after successful OTP validation.

REQ-REG-6: System shall provide an admin review interface for manual validation of patient identity documents.

REQ-REG-7: System shall allow administrators to approve or reject patient registrations with optional rejection reasons..

REQ-REG-8: System shall send an OTP to the patient's email for ownership verification.

REQ-REG-9: System shall require correct OTP entry before account creation.

REQ-REG-10: System shall enforce uniqueness of phone number and national ID.

REQ-REG-11: System shall hash and securely store passwords.

REQ-REG-12: System shall generate a UUID for each patient upon successful verification.

REQ-REG-13: System shall mark verified accounts as "Verified Patient."

REQ-REG-14: System shall log all identity-verification steps.

REQ-REG-15: System shall prevent unverified accounts from accessing medical-record features.

3.2 Patient Authentication

3.2.1 Description and Priority

Description: Enables patients to securely access their accounts using phone/email + password, with optional MFA and secure session handling. Includes mechanisms for account recovery via email or phone.

Priority: High

3.2.2 Stimulus/Response Sequences

1. Patient opens login page → system displays fields for phone/email and password.
2. Patients enters credentials → system validates format.
3. System checks if account exist → if not found return “invalid credentials”
4. System compares password hash → if incorrect return “invalid credentials”
5. If authentication succeeds → system generates a secure session token
6. System logs login timestamp and returns access token
- Optional MFA (if enabled)**
7. After password validation, system sends OTP to phone/email.
8. Patient enters OTP.
9. System validates OTP → grants access.
- Account Recovery (Forgot Password)**
10. Patient selects “Forgot Password”.
11. System asks for phone or email.
12. System sends OTP.
13. Patient enters OTP → system verifies.
14. System allows password reset.

3.2.3 Functional Requirements

- REQ-AUTH-1:** System shall allow patients to authenticate using either phone number or email plus password.
- REQ-AUTH-2:** System shall validate credentials against stored hashed passwords.
- REQ-AUTH-3:** System shall return a secure session token upon successful login.
- REQ-AUTH-4:** System shall record login timestamps.
- REQ-AUTH-5:** System shall reject login attempts using invalid phone/email formats.
- REQ-AUTH-6:** System shall reject invalid credentials without revealing which field was wrong.
- REQ-AUTH-7:** System shall support optional multi-factor authentication using OTP via email.
- REQ-AUTH-8:** System shall provide a password recovery flow using OTP verification.
- REQ-AUTH-9:** System shall allow password changes only after OTP verification.
- REQ-AUTH-10:** System shall lock an account temporarily after a configurable number of failed attempts (rate limiting + logout).
- REQ-AUTH-11:** System shall invalidate all active sessions after a password reset.
- REQ-AUTH-12:** System shall log all authentication-related events for auditing.
- REQ-AUTH-13:** System shall temporarily lock an account after a configurable number of failed login attempts (rate limiting + logout).

3.3 Patient Profile Management

3.3.1 Description and Priority

Description: This feature allows patients to view and update personal information required for identification, communication, and care coordination. Certain fields are editable by the patient, while others are locked and can only be modified by authorized healthcare personnel.

Priority: High

3.3.2 Stimulus/Response Sequences

1. Patient logs in → system displays dashboard with “View Profile” button.
2. Patient clicks “View Profile” → system shows all personal and medical baseline information.
3. Patient edits allowed fields (e.g., phone, email, address, portrait) → system validates input formats.
4. Patient submits updates → system stores changes securely, updates last updated timestamp.
5. Patient chooses to change password → system validates current password and applies new hashed password.
6. System can generate/re-generate a unique QR code for the patient.
7. Patient optionally requests account deletion → system checks policy and either processes or denies request.
8. System logs all profile-related actions (view/edit/password change/QR code generation/deletion attempt).

3.3.3 Functional Requirements

REQ-PROF-1: System shall display all personal and medical baseline data to the patient.

REQ-PROF-2: System shall allow editing of optional fields: phone, email, address, portrait photo, emergency contacts.

REQ-PROF-3: System shall prevent editing of read-only fields: national ID, date of birth, blood type (if mandated).

REQ-PROF-4: System shall validate all updated fields for correct format (e.g., phone number, email).

REQ-PROF-5: System shall store last updated timestamp for any profile changes.

REQ-PROF-6: System shall allow patients to change their password after verifying the current password.

REQ-PROF-7: System shall generate or regenerate a unique QR code for the patient.

REQ-PROF-8: System shall log all profile management actions for auditing purposes.

3.4 Patient Medical Records Access

3.4.1 Description and Priority

Description: Enables patients to view, filter, search, and securely share their medical records. All records are read only from the patient perspective. Patients can grant temporary access to healthcare providers via OTP, and download/export selected records in secure formats (CSV/PDF). Includes audit logging for all record-related actions.

Priority: High

3.4.2 Stimulus/Response Sequences

1. Patient logs in → navigates to “Medical Records” section.
2. System displays a chronological list of all medical records: visits, prescriptions, lab results, imaging reports, and emergency notes.
3. Patient selects a record → system displays full details of the record.
4. Patient applies filters/search (by date, record type, provider) → system updates displayed records accordingly.
5. Patient chooses to download/export a single record or multiple records → system generates a secure file (CSV or PDF).
6. Patient chooses to grant temporary access to a healthcare provider → system generates a time-limited OTP for the provider.
7. System logs all actions: viewed records, downloaded/exported records, shared OTPs.

3.4.3 Functional Requirements

REQ-MR-1: System shall allow patients to view all historical and current medical records in a read-only format.

REQ-MR-2: System shall display record details including visit date, symptoms, provider, diagnoses (ICD-11 codes), prescriptions, lab results, imaging summaries, and emergency notes.

REQ-MR-3: System shall allow filtering and searching of records by date, type, and provider.

REQ-MR-4: System shall prevent patients from editing or deleting any medical record.

REQ-MR-5: System shall allow patients to download/export records in secure formats (CSV, PDF).

REQ-MR-6: System shall allow patients to grant temporary access to their records to a specific healthcare provider by generating an OTP.

REQ-MR-7: System shall ensure OTPs for record access have a limited validity period and single-use restriction.

REQ-MR-8: System shall display attached lab files and imaging reports in a previewable format if applicable.

REQ-MR-9: System shall log all patient actions related to medical records (view, download/export, OTP generation) for auditing purposes.

REQ-MR-10: System shall notify the patient if an OTP grant is successfully created or fails

3.5 Patient Notifications & Reminders

3.5.1 Description and Priority

Description: Enables patients to receive notifications and reminders for scheduled appointments and prescribed medications, all of which are set by healthcare

providers. Patients can view notifications, mark them as read, and receive real-time alerts via mobile push notifications or in-app messages.

Priority: Medium-High

3.5.2 Stimulus/Response Sequences

1. Healthcare provider schedules an appointment or prescribes medication → system creates a reminder/notification for the patient.
2. Patient logs in or opens mobile app → system displays all active notifications and reminders.
3. Patient views a notification → system marks it as read.
4. System sends push notification at the scheduled time.
5. Patient interacts with notification (acknowledge or dismiss) → system updates status.

3.5.3 Functional Requirements

REQ-REM-1: System shall allow healthcare providers to create reminders/notifications for appointments and medications.

REQ-REM-2: System shall display all active notifications and reminders to the patient in chronological order.

REQ-REM-3: System shall allow patients to mark notifications as read.

REQ-REM-4: System shall send push notifications to mobile devices at the scheduled time.

REQ-REM-5: System shall log all notification-related events (creation, delivery, reading) for auditing.

REQ-REM-6: System shall prevent patients from creating, editing, or deleting reminders themselves.

3.6 Patient Data Access Logs

3.6.1 Description and Priority

Description: Allows patients to view a history of all entities that accessed their medical records. Each log entry includes the accessing entity (healthcare provider, lab, imaging center), the access type (read/write), and timestamp. Logs are immutable and cannot be edited or deleted by anyone, including administrators.

Priority: High

3.6.2 Stimulus/Response Sequences

Patient opens “Access Logs” section → system retrieves all relevant log entries.

System displays logs in chronological order with filters (date range, provider type optional).

Patient views a log entry → system shows its details (who accessed, what was accessed, when).

3.6.3 Functional Requirements

REQ-LOG-1: System shall record a log entry every time a patient's medical record is accessed.

REQ-LOG-2: Each log entry shall include:

- Accessing entity ID
- Access type (Read/Write)
- Timestamp

REQ-LOG-3: System shall prevent modification or deletion of log entries after creation.

REQ-LOG-4: System shall allow patients to view all access logs through the portal.

REQ-LOG-5: System shall present logs in reverse chronological order.

REQ-LOG-6: System shall ensure logs are tied to the correct patient via UUID.

REQ-LOG-7: System shall store logs in an append-only (immutable) manner.

3.7 Patient Notes

3.7.1 Description and Priority

Description: Allows patients to create and manage personal notes related to their health, symptoms, or daily observations. These notes are strictly patient-generated data and are not treated as medical records. Healthcare providers may view them but cannot edit them. Notes exist only to help patients remember details during consultations and to optionally give providers additional context.

Priority: Low–Medium

3.7.2 Stimulus/Response Sequences

1. Patient opens Notes → system shows existing notes.
2. Patient creates a new note → system saves it with a timestamp.
3. Patient views, edits, or deletes a note.
4. Provider views patient notes in read-only mode (logged).

3.7.3 Functional Requirements

REQ-NOTE-1: System shall allow patients to create, edit, view, and delete notes.

REQ-NOTE-2: System shall store notes separately from medical records.

REQ-NOTE-3: System shall timestamp each note.

REQ-NOTE-4: System shall restrict modification rights to the patient only.

REQ-NOTE-5: System shall allow providers to view notes in read-only mode.

REQ-NOTE-6: System shall log provider access to patient notes.

3.8 HCP Registration

3.8.1 Description and Priority

Description: Enables healthcare providers (HCPs) to create a professionally verified account on Sijill. Registration requires personal information, medical license details, and upload of identity and license documents. The system performs OCR on the medical license, validates license number format, cross-checks personal info with the uploaded ID, and optionally performs face-matching. Because external APIs may not be available (student implementation), the system uses automated document verification combined with manual approval to ensure only legitimate healthcare providers can access patient records. Unverified HCPs cannot view or create patient encounters.

Priority: High

3.8.2 Stimulus/Response Sequences

1. HCP opens registration page → system displays registration form.
2. HCP enters **mandatory** information:
 - Full Name (First/Middle/Surname)
 - Date of birth
 - Gender
 - Email
 - Specialization
 - Medical License Number
 - Practice Address / Workplace
3. System requests professional identity verification → HCP must upload:
 - National ID (Front + Back)
 - Medical License Document (Scan or Photo)
 - Optional Workplace Document
4. System performs automated checks:
 - OCR extracts information from license: name, license number, specialization, expiry date
 - Validates that license number follows expected format
 - Matches name across license, ID, and registration form

- *Validates that license expiry date is valid*
 - *face-matching between ID photo and profile photo*
6. *System performs email verification via OTP:*
- *Correct → continue*
 - *Incorrect → retry*
7. *If all automated checks pass → HCP account enters **Pending Verification** state.*
8. *An authorized system administrator (university demo: project team) reviews documents and approves/rejects.*
9. *Upon approval → system marks HCP as **Verified HCP**, generates UUID, and enables access.*
10. *System confirms successful activation and provides login option.*

3.8.3 Functional Requirements

REQ-HCP-REG-1: *System shall require the following mandatory fields: full name, date of birth, gender, phone number, specialization, medical license number, practice address.*

REQ-HCP-REG-2: *System shall allow optional fields: email, profile photo.*

REQ-HCP-REG-3: *System shall require upload of medical license document and National ID.*

REQ-HCP-REG-4: *System shall perform OCR extraction on medical license to read license number, name, specialization, and expiry date.*

REQ-HCP-REG-5: *System shall validate that the license number follows expected structure and format.*

REQ-HCP-REG-6: *System shall match the name extracted from license with ID and registration data.*

REQ-HCP-REG-7: *System shall validate that the medical license is not expired.*

REQ-HCP-REG-8: *System shall optionally perform face-matching between ID photo and profile photo.*

REQ-HCP-REG-9: *System shall send OTP to the HCP email and require correct entry.*

REQ-HCP-REG-10: *System shall enforce uniqueness of email across all HCP accounts.*

REQ-HCP-REG-11: *System shall hash and securely store passwords.*

REQ-HCP-REG-12: *System shall generate a UUID for each HCP upon approval.*

REQ-HCP-REG-13: *System shall assign HCP accounts one of the following statuses: Pending Verification, Verified, or Rejected.*

REQ-HCP-REG-14: *System shall prevent unverified HCP accounts from accessing patient data or creating encounters.*

REQ-HCP-REG-15: *System shall notify HCPs of approval or rejection.*

REQ-HCP-REG-16: *System shall log all document verification and approval actions.*

3.9 HCP Authentication

3.9.1 Description and Priority

Description: Enables doctors to securely access their accounts using phone number and password. The system supports optional multi-factor authentication (MFA) via OTP for added security. Email can be used for account recovery. Authentication ensures only verified doctors can access patient data and perform actions.

Priority: High

3.9.2 Stimulus/Response Sequences

1- Doctor opens the login page → system displays fields for phone number and password.

2- Doctor enters credentials → system validates the format of the phone number.

3- System checks if an account exists for the provided phone number:

- Not found → system returns “invalid credentials.”
- Found → system compares password hash.

4- Password validation:

- Incorrect → system returns “invalid credentials.”
- Correct → proceed to optional MFA.

5- Optional MFA via OTP:

- System sends OTP to doctor’s email.
- Doctor enters OTP → system validates:
 - Correct → login successful.
 - Incorrect → retry or lockout after configurable attempts.

6- System generates a secure session token, logs login timestamp, and grants access.

Account Recovery (Forgot Password)

7. Doctor selects “Forgot Password.”

8. System prompts for registered email.

9. System sends OTP to phone/email.

10. Doctor enters OTP → system verifies.

11. Doctor sets a new password → system hashes and stores it securely.

12. System invalidates all previous sessions for security.

3.9.3 Functional Requirements

REQ-HCP-AUTH-1: System shall allow doctors to authenticate using phone number and password.

REQ-HCP-AUTH-2: System shall validate credentials against stored hashed passwords.

REQ-HCP-AUTH-3: System shall return a secure session token upon successful login.

REQ-HCP-AUTH-4: System shall log login timestamps and IP addresses for auditing.

REQ-HCP-AUTH-5: System shall reject login attempts using invalid phone number formats.

REQ-HCP-AUTH-8: System shall support optional multi-factor authentication (MFA) using OTP via email.

REQ-HCP-AUTH-9: System shall provide a password recovery flow using OTP verification.

REQ-HCP-AUTH-10: System shall allow password changes only after OTP verification.

REQ-HCP-AUTH-11: System shall temporarily lock an account after a configurable number of failed login attempts (rate limiting + logout).

REQ-HCP-AUTH-12: System shall invalidate all active sessions after a password reset.

REQ-HCP-AUTH-13: System shall log all authentication-related events for auditing purposes.

3.10 HCP Profile Management

3.10.1 Description and Priority

Description: Doctors can view their profile and update editable fields (phone, email, profile photo, specialization). Critical fields (name, DOB, license number) are read-only. All changes are validated and logged.

Priority: High

3.10.2 Stimulus/Response Sequences

1- Doctor logs in → system displays the dashboard with a “View Profile” option.

2- Doctor selects “View Profile” → system displays all personal and professional data.

3- Doctor edits allowed fields (phone number, email, profile photo, specialization).

4- Doctor submits updates → system validates input formats:

- Phone number format
- Email format

5- If validation passes → system stores changes securely and updates the last modified timestamp.

6- If validation fails → system notifies doctor with specific error messages and does not save changes.

7- System logs all profile-related actions including view and edit events for auditing purposes.

3.10.3 Functional Requirements

REQ-HCP-PROF-1: System shall allow doctors to view their full profile.

REQ-HCP-PROF-2: System shall allow editing of phone number, email, profile photo, and specialization.

REQ-HCP-PROF-3: System shall prevent editing of read-only fields: full name, date of birth, medical license number.

REQ-HCP-PROF-4: System shall validate updated phone numbers and email addresses for correct format.

REQ-HCP-PROF-5: System shall store a timestamp for any profile changes.

3.11 HCP Medical Record Management

3.11.1 Description and Priority

Description: This feature allows Healthcare Providers (HCPs) to access a patient's full medical history after obtaining patient-granted, time-limited access via a One-Time Password (OTP). Once access is granted, the HCP can create new clinical encounter records. Each encounter record documents a single patient visit and includes structured and unstructured data for symptoms (with SNOMED CT codes and optional image attachments), diagnoses (with ICD-11 codes), prescribed medications, lab/imaging orders, and treatment notes. The system automatically generates patient reminders for medications and upcoming appointments based on the data entered, and creates pending test orders in the patient's portal. HCPs can also export a summary of the current encounter as a PDF.

Priority: High

3.11.2 Stimulus/Response Sequences

1. HCP navigates to their dashboard and selects “New Patient Report”
2. System prompts HCP to enter the OTP generated from the patient
3. The patient opens the APP and navigate to the “Grant Access” option and generates the OTP → HCP enters OTP
4. System validates the OTP → if valid, grants access to the patient medical history and access to create a new record (for a configurable duration)
5. HCP selects “View Medical History” → system lists a chronological list of all the records in the patient medical history
6. HCP selects “Add New Encounter” → system presents a structured encounter from
 - **Symptoms:** HCP searches SNOMED CT via a search box → selects relevant symptoms → also writes symptoms details (free textbox) → can also attach image files to specific symptoms
 - **Diagnosis:** HCP searches ICD-11 via a search box → selects one or more codes → system auto populates diagnosis name and descriptions in which the HCP can edit
 - **Medications:** HCP enters medication name, dosage, form (dropdown), frequency, duration (start date → end date), and instructions (all as free textbox except form) → submitting this automatically creates a medication reminder for the patient
 - **Orders:** HCP requests a Lab/Imaging tests if needed → system creates an appending order in the patient portal
 - HCP sets a date for the next appointment → system creates an appointment reminder for the patient
7. HCP submits the completed encounter form → System saves the record, logs the action, and makes the newly created reminders/orders active for the patient.
8. HCP can select "Export as PDF" for any encounter → System generates a PDF summary of that single encounter and downloads it to the HCP's device.

3.11.3 Functional Requirements

REQ-HCP-MRM-1: The system shall require a valid, patient-granted OTP before allowing an HCP to create a new encounter record or access a patient's full medical history.

REQ-HCP-MRM-2: The system shall provide a form for creating a new encounter record, associated with a specific patient and the HCP creating it.

REQ-HCP-MRM-3: For documenting symptoms, the system shall provide a searchable interface to the SNOMED CT terminology (via a public API) allowing multiple selections.

REQ-HCP-MRM-4: The system shall allow an HCP to add free-text details to any selected SNOMED CT symptom.

REQ-HCP-MRM-5: The system shall allow an HCP to attach one or more image files (e.g., .jpg, .png) to any specific symptom entry.

REQ-HCP-MRM-6: For documenting diagnoses, the system shall provide a searchable interface to the ICD-11 terminology (via a public API) allowing multiple selections.

REQ-HCP-MRM-7: The system shall auto-populate the diagnosis description from ICD-11 but allow the HCP to override this text for clinical accuracy.

REQ-HCP-MRM-8: The system shall provide structured fields for prescribing medication, including: Medication Name (free-text), Dosage (free-text), Form (dropdown), Frequency (dropdown with "Other" free-text option), Duration (free-text), and Instructions (free-text).

REQ-HCP-MRM-9: Upon saving an encounter with prescribed medications, the system shall automatically create corresponding medication reminders in the patient's portal.

REQ-HCP-MRM-10: The system shall provide a field for entering lab or imaging test orders as free-text descriptions.

REQ-HCP-MRM-11: Upon saving an encounter with test orders, the system shall create a corresponding pending order in the patient's portal, which the patient can use to generate an OTP for the laboratory or imaging center.

REQ-HCP-MRM-12: The system shall provide a field for setting the next appointment date. Upon saving the encounter, the system shall automatically create an appointment reminder for the patient.

REQ-HCP-MRM-13: The system shall allow an HCP to export a summary of a single encounter record as a PDF file. The export function shall not allow the export of a patient's full historical medical history.

REQ-HCP-MRM-14: The system shall log the creation of every encounter record and all subsequent views of it for audit purposes.

3.12 HCP Patients Profiles & Encounter History

3.12.1 Description and Priority

Description: This feature provides Healthcare Providers (HCPs) with a dedicated list of all patients with whom they have previously created at least one clinical encounter. For each patient, the HCP can view the patient's shareable profile and a chronological history of all encounters performed specifically by that HCP. This allows the HCP to quickly review past interactions and patient baseline information without requiring a new OTP for each access.

Priority: High

3.12.2 Stimulus/Response Sequences

1. HCP navigates to their dashboard and selects "My Patients".
2. System displays a list of patient profiles, showing for each: Patient Name, Date of Last Encounter, and the Primary Diagnosis from the last encounter.
3. HCP selects a specific patient from the list.

4. System displays a detailed view split into two sections:
 - a. Patient Profile: Displays the patient's shareable information (Full Name, Gender, Date of Birth, Blood Type, Portrait).
 - b. Encounter History: A list of all clinical encounters the current HCP has created for this patient, showing the date and primary diagnosis for each.
5. HCP selects a specific past encounter from the history list → System displays the full details of that encounter record in a read-only view.

3.12.3 Functional Requirements

REQ-HCP-PATLIST-1: The system shall maintain and display a list for each HCP containing all patients for whom the HCP has created at least one encounter record.

REQ-HCP-PATLIST-2: For each patient in the list, the system shall display the patient's name, the date of the last encounter, and the primary diagnosis from that encounter.

REQ-HCP-PATLIST-3: Upon selecting a patient, the system shall display the patient's shareable profile information (Full Name, Gender, Date of Birth, Blood Type, Portrait).

REQ-HCP-PATLIST-4: The system shall display a chronological list of all clinical encounters created for the selected patient by the current HCP.

REQ-HCP-PATLIST-5: The system shall allow the HCP to view the full details of any encounter in the history list in a read-only format.

3.13 Patient Active Diagnosis & Outcome Tracking

3.13.1 Description and Priority

Description: This feature maintains a dynamic, patient-level list of "Active Diagnoses" that provides a continuous overview of a patient's unresolved health conditions. Diagnoses are automatically added to this list when a Healthcare Provider (HCP) creates a clinical encounter and marks a diagnosis as a new, active problem. Diagnoses on this list can be resolved in two ways: manually by an HCP during a subsequent clinical encounter, or automatically when the patient self-reports a successful outcome via an automated, system-generated follow-up prompt. This ensures all diagnoses are tracked to resolution, providing valuable data for research and patient care.

Priority: Medium-High

3.13.2 Stimulus/Response Sequences

1- During the creation of an encounter (Feature 3.11), when an HCP adds a diagnosis, the system automatically adds this diagnosis to the active diagnosis list of the patient

Viewing the Active Diagnoses List:

2- Patients view their own "Active Diagnoses" list in their portal.

3- An HCP with access to a patient's record can view the patient's "Active Diagnoses" list as part of their profile or medical history.

Resolving a Diagnosis via HCP (During a New Encounter):

4- During a new, legitimate clinical encounter, an HCP with access views the patient's "Active Diagnoses" list.

5- The HCP selects an option to "Resolve" a specific diagnosis from the list.

- 6- The system prompts the HCP to optionally add a note
- 7- The HCP confirms → The system removes the diagnosis from the "Active Diagnoses" list, marks it as "Resolved by HCP" with a timestamp.

Resolving a Diagnosis via Automated Patient Follow-up:

- 8- After a configurable global delay (e.g., 7 days) from the date a diagnosis was added to the "Active Diagnoses" list, the system automatically sends a push notification to the patient's mobile app.

- 9- Patient opens the notification → System presents a structured question: "Regarding your diagnosis of [Diagnosis Name], how are you feeling now?"

- 10- Patient selects one response from: Fully Recovered, Improved, No Change, Worse. An optional free-text field for details is provided.

- 11- Patient submits the response.

* If the response is Fully Recovered, the system automatically removes the diagnosis from the "Active Diagnoses" list, marks it as "Resolved by Patient" and records the outcome.

* If the response is Improved or No Change or Worse, the system records the response and schedules a new follow-up prompt after another configured interval (e.g., another 7 days), without removing the diagnosis from the active list.

3.13.3 Functional Requirements

REQ-DX-1: The system shall maintain a patient-specific "Active Diagnoses" list, distinct from encounter records.

REQ-DX-2: In an encounter when an HCP adds a new diagnosis, the system shall automatically add the diagnosis to the patient's "Active Diagnoses" list.

REQ-DX-3: The system shall allow an HCP with record access to resolve any diagnosis on the "Active Diagnoses" list during a new clinical encounter, logging the resolution within that new encounter.

REQ-DX-4: The system shall automatically initiate a follow-up process for any diagnosis on the "Active Diagnoses" list after a globally configurable time delay.

REQ-DX-5: The automated follow-up shall prompt the patient to report their outcome with structured responses (Fully Recovered, Improved, No Change, Worse).

REQ-DX-6: Upon a patient reporting Fully Recovered or Worse, the system shall automatically resolve the diagnosis on the active list and record the patient's outcome.

REQ-DX-7: Upon a patient reporting Improved or No Change, the system shall record the response and re-initiate the follow-up process after a subsequent configured interval.

REQ-DX-8: The system shall display the current "Active Diagnoses" list and the resolution history (who resolved it and when) to authorized users (Patient and HCPs with access).

3.14 Test Lab Registration

3.14.1 Description and Priority

Description: Enables diagnostic laboratories to register a single organization-level account with Sijill. A lab center account represents the organization (center/branch), not individual technicians. Registration requires center identity documents (company/trade license or lab accreditation), a primary contact phone number, and an administrator contact email. The system performs automated document ingestion (OCR) and basic validation, places the

account into Pending Verification, and requires an authorized approver (project demo: team member) to mark the lab as Verified before the center can authenticate and submit results. Unverified centers cannot upload results or access any patient-related features.

Priority: High

3.14.2 Stimulus/Response Sequences

1. Lab admin opens "Register Lab Center" page → system displays registration form.

2. Lab admin provides mandatory information:

- Center Name*
- National/Commercial Registration Number or Accreditation Number*
- Address*
- Administrator Full Name and Contact Phone*
- Administrator Email*

3. Lab admin uploads required documents:

- Official lab accreditation / business registration (scan/photo, PDF)*
- Proof of address (optional)*
- Center logo (optional)*

4. System validates input formats and document presence.

5. System performs automated processing: OCR on uploaded documents to extract registration/accreditation number and center name.

6. System sends OTP to the primary email → lab admin enters OTP:

- Correct → continue*
- Incorrect → retry up to configurable attempts*

7. System creates lab center account in Pending Verification state and stores all data securely.

8. Authorized approver reviews the uploaded documents and OCR results. Approver either Approves → account becomes Verified, or Rejects → account becomes Rejected with reason.

9. System notifies the lab admin of approval or rejection. Approved center receives credentials / activation instructions.

10. Once Verified, the lab center can authenticate and use result-submission features.

3.14.3 Functional Requirements

REQ-LAB-REG-1: System shall require the following mandatory fields: center name, registration/accreditation number, primary phone number, administrator name, administrator phone, administrator email.

REQ-LAB-REG-2: System shall allow optional fields: center address, center logo, supporting documents.

REQ-LAB-REG-3: System shall require upload of at least one official document proving the center's registration or accreditation.

REQ-LAB-REG-4: System shall perform OCR on uploaded documents and extract registration/accreditation number and center name.

REQ-LAB-REG-5: System shall send an OTP to the center's primary phone number and require OTP verification to proceed.

REQ-LAB-REG-6: System shall create accounts in a Pending Verification state after automated checks.

REQ-LAB-REG-7: System shall provide an admin review queue for manual approval or rejection of pending centers.

REQ-LAB-REG-8: System shall mark accounts with one of three statuses: Pending Verification, Verified, Rejected.

REQ-LAB-REG-9: System shall prevent Pending or Rejected centers from authenticating to result-submission endpoints.

REQ-LAB-REG-10: System shall enforce uniqueness of the center registration/accreditation number.

REQ-LAB-REG-11: System shall store timestamps for registration submission, verification actions, and status changes.

REQ-LAB-REG-12: System shall log all registration and verification actions in append-only audit logs.

REQ-LAB-REG-13: System shall store uploaded documents encrypted at rest and restrict access to authorized approvers.

REQ-LAB-REG-14: System shall notify the lab admin by email/SMS when the account is approved or rejected, including rejection reason.

3.15 Test Lab Authentication

3.15.1 Description and Priority

Description: Enables verified lab centers to authenticate and access the limited set of actions required for submission result. Authentication supports username (center ID or admin email) + password, with optional MFA (OTP). Only Verified lab centers may authenticate. Sessions are narrowly scoped and short-lived. All authentication events are audited.

Priority: High

3.15.2 Stimulus/Response Sequences

1. Lab admin opens login page → system displays fields for center identifier (center ID or admin email) and password.
2. Lab admin enters credentials → system validates format.
3. If credentials match a Verified center account → system proceeds to optional
 - If MFA enabled → system sends OTP to primary center phone or admin email → lab admin enters OTP → system validates.
4. Upon successful authentication → system issues a short-lived session token with scope limited to result-submission APIs.
5. System logs the authentication event (center id, IP, timestamp).
6. If authentication fails → system returns an error without revealing which field was incorrect and increments rate-limiting counters.

3.15.3 Functional Requirements

REQ-LAB-AUTH-1: System shall allow lab centers to authenticate using center ID (or admin email) and password.

REQ-LAB-AUTH-2: System shall validate passwords against securely stored hashes.

REQ-LAB-AUTH-3: System shall permit optional MFA via OTP sent to the center's registered email.

REQ-LAB-AUTH-4: System shall only allow authentication for accounts in **Verified** status.

REQ-LAB-AUTH-5: System shall generate a short-lived session token scoped to result submission APIs upon successful authentication.

REQ-LAB-AUTH-6: System shall log all authentication attempts with timestamp, IP, and result.

REQ-LAB-AUTH-7: System shall implement rate limiting and temporary lockout after configurable failed attempts.

REQ-LAB-AUTH-8: System shall invalidate session tokens after a configurable idle timeout or manual logout.

REQ-LAB-AUTH-9: System shall prevent session tokens from granting broader access than upload-only operations.

3.16 Test Lab Result Submission

3.16.1 Description and Priority

Description: Defines the secure, upload-only workflow used by lab centers to submit test results. Access is granted either via: (A) Doctor-initiated order (Order Token + order-linked OTP generated by the patient) which attaches results to a specific encounter, or (B) Patient-initiated diagnostic (Patient-generated OTP for upload-only access) which creates a self-requested diagnostic entry. Lab centers authenticate, present an Order Token or Patient OTP, and may only upload structured results and/or supporting PDF attachments. Centers cannot view patient history, export data, or modify previously submitted results. All uploads are stored immutably and are visible to the patient and any authorized HCPs (if an order was present).

Priority: High

3.16.2 Stimulus/Response Sequences

A. Doctor-Initiated Order Flow

1. Doctor creates lab order in the patient's encounter → system generates a unique, single-use Order Token and stores order metadata (order id, tests, ordering doctor, expiry).
2. Patient sees the order in their app and chooses Generate OTP for Lab → system sends OTP to patient phone and associates OTP to the Order Token.
3. Lab center authenticates → enters Order Token and patient OTP.
4. System validates: Order Token exists and is not expired; OTP matches and is single-use.
5. If validation passes → system opens an Upload Session for this order (upload-only, scoped).
6. Lab center uploads: structured test results (JSON form per test), optional PDF report, optional attachments (image scans).
7. Lab center submits results → system validates required fields and stores the result as a completed order, sets order status = Completed, and attaches result to the specific encounter and patient record.
8. System notifies ordering doctor and patient, logs the action, and closes the Upload Session.

B. Patient-Initiated (Self-Requested) Flow

1. Patient requests lab access from their app via "Give Lab Access" → system generates a single-use Patient OTP scoped to upload-only and valid for a short interval.
2. Lab center authenticates → enters Patient OTP.
3. System validates OTP (exists, valid, not expired).
4. If valid → system opens an Upload Session for a Self-Requested Diagnostic Encounter.
5. Lab center uploads structured results, optional PDF.
6. Upon submission → system creates a Self-Requested Diagnostic encounter in the patient's record, stores results, logs actions, notifies patient, and closes the
7. Upload Session.

Common rules for both flows

- *Upload Session expires immediately after successful submission or after configurable timeout (e.g., 60 minutes) if incomplete.*
- *Lab center cannot list or query patients by name or national ID; only Order Token or Patient OTP may open an Upload Session.*
- *All uploads are scanned/validated for allowed file types and size limits; malicious file detection runs before storage.*
- *All actions are logged immutably.*

3.16.3 Functional Requirements

REQ-LAB-UP-1: System shall accept two submission flows: Doctor-Initiated (Order Token + Order-linked OTP) and Patient-Initiated (Patient OTP).

REQ-LAB-UP-2: System shall generate a unique Order Token for each doctor-created order and store its expiry and metadata.

REQ-LAB-UP-3: System shall allow patients to generate an OTP per order (attaching OTP to the order) and to generate an OTP for self-requested diagnostics.

REQ-LAB-UP-4: System shall validate Order Token and OTP pair before opening an Upload Session.

REQ-LAB-UP-5: System shall restrict Upload Sessions to upload-only operations and prevent any read access to patient history.

REQ-LAB-UP-6: System shall allow lab centers to upload structured lab results per a standardized schema (see below) and optional PDF attachments.

REQ-LAB-UP-7: System shall not allow centers to export patient data; downloads of the submitted result by centers are prohibited.

REQ-LAB-UP-8: System shall validate file types (PDF, JPEG/PNG for images, ZIP for bulk attachments) and maximum file sizes before accepting uploads.

REQ-LAB-UP-9: System shall scan uploaded files for malware and reject suspicious uploads.

REQ-LAB-UP-10: System shall attach completed lab results to the correct patient record and to the ordering encounter if the Order Token belonged to an order.

REQ-LAB-UP-11: System shall create a Self-Requested Diagnostic encounter when results are uploaded via Patient OTP with no associated order.

REQ-LAB-UP-12: System shall set the Order Token/OTP to single-use and mark it expired after successful submission.

REQ-LAB-UP-13: System shall close Upload Sessions automatically after successful submission or after a configurable idle timeout (default 60 minutes).

REQ-LAB-UP-14: System shall log all Upload Session events (token used, center id, uploader id, timestamps, IP) in append-only audit logs.

REQ-LAB-UP-15: System shall notify the patient and the ordering HCP (if any) upon successful result submission.

REQ-LAB-UP-16: System shall keep uploaded results immutable once submitted; corrections require a new report and are versioned.

REQ-LAB-UP-17: System shall store structured lab results in a standardized schema that includes at minimum: test code (LOINC or internal), test name, result value, unit, reference range, result flag (normal/low/high), sample collection timestamp, result timestamp, performing center id, and optional technician notes.

REQ-LAB-UP-18: System shall encrypt uploaded files at rest and use HTTPS/TLS for transport.

REQ-LAB-UP-19: System shall enforce rate limits and abuse protections on

token/OTP validation endpoints.

REQ-LAB-UP-20: *System shall support retraction/replacement procedure: if a lab must correct a submitted result, the center may upload a corrected report flagged as “amendment” and the system shall retain both versions with clear timestamps and user IDs.*

3.17 Imagine Center Registration

3.17.1 Description and Priority

Description: *Enables diagnostic imaging centers (e.g., Radiology, MRI, CT scan centers) to register a single organization-level account with Sijill. This account represents the imaging center as an organization, not individual radiologists or technicians. Registration requires center identity documents (business registration, radiology accreditation), a primary contact phone number, and an administrator contact email. The system performs automated document ingestion (OCR) and basic validation, places the account into a Pending Verification state, and requires an authorized approver (project demo: team member) to mark the center as Verified before it can authenticate and submit imaging studies. Unverified centers cannot upload studies or access any patient-related features.*

Priority: *High*

3.17.2 Stimulus/Response Sequences

1. *Imaging Center admin opens "Register Imaging Center" page → system displays registration form.*
2. *Imaging Center admin provides mandatory information:*
 - *Center Name*
 - *National/Commercial Registration Number or Radiology Accreditation Number*
 - *Address*
 - *Administrator Full Name and Contact Phone*
 - *Administrator Email*
3. *Imaging Center admin uploads required documents:*
 - *Official radiology accreditation / business registration (scan/photo, PDF)*
 - *Proof of address (optional)*
 - *Center logo (optional)*
4. *System validates input formats and document presence.*

5. *System performs automated processing: OCR on uploaded documents to extract registration/accreditation number and center name.*
6. *System sends OTP to the primary phone → Imaging Center admin enters OTP:*
 - *Correct → continue*
 - *Incorrect → retry up to configurable attempts*
7. *System creates imaging center account in Pending Verification state and stores all data securely.*
8. *Authorized approver reviews the uploaded documents and OCR results. Approver either Approves → account becomes Verified, or Rejects → account becomes Rejected with a reason.*
9. *System notifies the Imaging Center admin of approval or rejection. An approved center receives credentials / activation instructions.*
10. *Once Verified, the imaging center can authenticate and use study-submission features.*

3.17.3 Functional Requirements

REQ-IMG-REG-1: *System shall require the following mandatory fields: center name, registration/accreditation number, primary phone number, administrator name, administrator phone, administrator email.*

REQ-IMG-REG-2: *System shall allow optional fields: center address, center logo, supporting documents.*

REQ-IMG-REG-3: *System shall require upload of at least one official document proving the center's registration or radiology accreditation.*

REQ-IMG-REG-4: *System shall perform OCR on uploaded documents and extract registration/accreditation number and center name.*

REQ-IMG-REG-5: *System shall send an OTP to the center's primary email and require OTP verification to proceed.*

REQ-IMG-REG-6: *System shall create accounts in a Pending Verification state after automated checks.*

REQ-IMG-REG-7: *System shall provide an admin review queue for manual approval or rejection of pending centers.*

REQ-IMG-REG-8: *System shall mark accounts with one of three statuses: Pending Verification, Verified, Rejected.*

REQ-IMG-REG-9: *System shall prevent Pending or Rejected centers from authenticating to study-submission endpoints.*

REQ-IMG-REG-10: *System shall enforce uniqueness of the center registration/accreditation number.*

REQ-IMG-REG-11: *System shall store timestamps for registration submission, verification actions, and status changes.*

REQ-IMG-REG-12: *System shall log all registration and verification actions in append-only audit logs.*

REQ-IMG-REG-13: *System shall store uploaded documents encrypted at rest and restrict access to authorized approvers.*

REQ-IMG-REG-14: *System shall notify the Imaging Center admin by email/SMS when the account is approved or rejected, including rejection reason.*

3.18 Imaging Center Authentication

3.18.1 Description and Priority

Description: Enables verified imaging centers to authenticate and access the limited set of actions required for study submission. Authentication supports a username (center ID or admin email) and password, with optional MFA (OTP). Only Verified imaging centers may authenticate. Sessions are narrowly scoped and short-lived. All authentication events are audited.

Priority: High

3.18.2 Stimulus/Response Sequences

1. Imaging Center admin opens login page → system displays fields for center identifier (center ID or admin email) and password.
2. Imaging Center admin enters credentials → system validates format.
3. If credentials match a Verified center account → system proceeds to optional MFA.
 - If MFA enabled → system sends OTP to primary center phone or admin email → Imaging Center admin enters OTP → system validates.
4. Upon successful authentication → system issues a short-lived session token with scope limited to study-submission APIs.
5. System logs the authentication event (center id, IP, timestamp).
6. If authentication fails → system returns an error without revealing which field was incorrect and increments rate-limiting counters.

3.18.3 Functional Requirements

REQ-IMG-AUTH-1: System shall allow imaging centers to authenticate using center ID (or admin email) and password.

REQ-IMG-AUTH-2: System shall validate passwords against securely stored hashes.

REQ-IMG-AUTH-3: System shall permit optional MFA via OTP sent to the center's registered admin email.

REQ-IMG-AUTH-4: System shall only allow authentication for accounts in Verified status.

REQ-IMG-AUTH-5: System shall generate a short-lived session token scoped to study submission APIs upon successful authentication.

REQ-IMG-AUTH-6: System shall log all authentication attempts with timestamp, IP, and result.

REQ-IMG-AUTH-7: System shall implement rate limiting and temporary lockout after configurable failed attempts.

REQ-IMG-AUTH-8: System shall invalidate session tokens after a configurable idle timeout or manual logout.

REQ-IMG-AUTH-9: System shall prevent session tokens from granting broader access than upload-only operations.

3.19 Imagin Center Result Submission

3.19.1 Description and Priority

Description: Defines the secure, upload-only workflow used by imaging centers to submit imaging studies and associated radiology reports. Access is granted either via: (A) Doctor-initiated order (Order Token + order-linked OTP generated by the patient) which attaches the study to a specific encounter, or (B) Patient-initiated diagnostic (Patient-generated OTP for upload-only access) which creates a self-requested diagnostic imaging entry. Imaging centers authenticate, present an Order Token or Patient OTP, and may upload DICOM files, standard image formats (JPEG, PNG), and/or a radiology report (PDF). Centers cannot view patient history, export data, or modify previously submitted studies. All uploads are stored immutably and are visible to the patient and any authorized HCPs.

Priority: High

3.19.2 Stimulus/Response Sequences

A. Doctor-Initiated Order Flow

1. Doctor creates an imaging order in the patient's encounter → system generates a unique, single-use Order Token and stores order metadata (order id, study type, ordering doctor, expiry).
2. Patient sees the order in their app and chooses "Generate OTP for Imaging" → system sends OTP to patient phone and associates OTP with the Order Token.
3. Imaging center authenticates → enters Order Token and patient OTP.
4. System validates: Order Token exists and is not expired; OTP matches and is single-use.
5. If validation passes → system opens an Upload Session for this order (upload-only, scoped).
6. Imaging center uploads: DICOM files and/or standard image files (JPEG/PNG), and an optional radiology report (PDF).
7. Imaging center submits the study → system validates file formats (ensures DICOM compliance) and stores the study as a completed order, sets order status = Completed, and attaches it to the specific encounter and patient record.
8. System notifies ordering doctor and patient, logs the action, and closes the Upload Session.

B. Patient-Initiated (Self-Requested) Flow

1. *Patient requests imaging center access from their app via "Give Imaging Center Access" → system generates a single-use Patient OTP scoped to upload-only and valid for a short interval.*
2. *Imaging center authenticates → enters Patient OTP.*
3. *System validates OTP (exists, valid, not expired).*
4. *If valid → system opens an Upload Session for a Self-Requested Diagnostic Imaging Encounter.*
5. *Imaging center uploads DICOM/standard image files and an optional radiology report (PDF).*
6. *Upon submission → system creates a Self-Requested Diagnostic Imaging encounter in the patient's record, stores the study, logs actions, notifies patient, and closes the Upload Session.*

Common rules for both flows

- *Upload Session expires immediately after successful submission or after a configurable timeout (e.g., 60 minutes) if incomplete.*
- *Imaging center cannot list or query patients by name or national ID; only an Order Token or Patient OTP may open an Upload Session.*
- *All uploads are scanned/validated for allowed file types (e.g., .dcm, .jpg, .png, .pdf) and size limits; malicious file detection runs before storage.*
- *All actions are logged immutably.*

3.19.3 Functional Requirements

REQ-IMG-SUB-1: *System shall allow a Verified imaging center to start a submission session only by providing a valid Order Token + Patient OTP (for ordered studies) or a valid Patient OTP (for self-requested studies).*

REQ-IMG-SUB-2: *System shall validate that an Order Token is active, unfulfilled, and matches the provided Patient OTP.*

REQ-IMG-SUB-3: *System shall validate that a Patient OTP for self-requested studies is valid, single-use, and has not expired.*

REQ-IMG-SUB-4: *System shall allow the upload of one or more DICOM files and/or standard image files (JPEG, PNG) per imaging study.*

REQ-IMG-SUB-5: *System shall allow the upload of a radiology report in PDF format.*

REQ-IMG-SUB-6: *System shall validate that uploaded DICOM files conform to the DICOM standard.*

REQ-IMG-SUB-7: *System shall scan all uploaded files for viruses and malicious content before permanent storage.*

REQ-IMG-SUB-8: *Upon successful submission for a doctor's order, the system shall link the imaging study to the original clinical encounter and mark the order as completed.*

REQ-IMG-SUB-9: *Upon successful submission for a self-requested study, the*

system shall create a new "Self-Requested Diagnostic Imaging" record in the patient's medical history.

REQ-IMG-SUB-10: *System shall prevent an imaging center from viewing, modifying, or deleting any previously submitted study.*

REQ-IMG-SUB-11: *System shall notify the patient and the ordering healthcare provider (if applicable) upon successful study submission.*

REQ-IMG-SUB-12: *System shall immutably log all submission session activities, including center ID, patient UUID, OTP used, timestamps, and file hashes.*

3.20 Super Admin Portal

3.20.1 Description and Priority

Description: The Super Admin Portal provides the highest level of system access for managing administrator accounts and overseeing critical system operations. Super admins can create and manage regular admin accounts, configure system-wide settings, and have full visibility into all administrative actions. This is a highly restricted portal with typically only 1-2 super admin accounts for the entire system.

Priority: High

3.20.2 Stimulus/Response Sequences

Admin Account Management:

1. *Super Admin logs in → system displays Super Admin dashboard*
2. *Super Admin navigates to "Admin Management" → system displays list of all admin accounts*
3. *Super Admin selects "Create New Admin"*
4. *System displays admin registration form:*
 - *Full Name*
 - *Email*
 - *Initial Password (system-generated)*
 - *Phone Number (optional)*
 - *Role assignment (currently only "Administrator")*
5. *Super Admin submits form → system creates admin account and sends credentials via email*
6. *System logs admin creation action with super admin ID and timestamp*
7. *Super Admin can view, suspend, or delete admin accounts → system logs all actions*

System Configuration:

8. Super Admin navigates to "System Settings"
9. System displays configurable parameters:
 - OTP expiry duration
 - Session timeout durations (per user type)
 - File upload size limits
 - Rate limiting thresholds
 - Automated follow-up intervals for diagnoses
10. Super Admin modifies settings → system validates and applies changes
11. System logs all configuration changes with before/after values

Audit Log Access:

12. Super Admin navigates to "Full Audit Logs"
13. System displays all system-wide logs with filtering options: - By user type (patient, HCP, lab, imaging, admin) - By action type (login, registration, verification, data access, etc.) - By date range - By specific user ID
14. Super Admin can export audit logs → system generates CSV/PDF report

3.20.3 Functional Requirements**Admin Management:**

REQ-SUPERADMIN-1: System shall restrict Super Admin accounts to a maximum of 2 accounts.

REQ-SUPERADMIN-2: System shall allow Super Admin to create new Administrator accounts.

REQ-SUPERADMIN-3: System shall generate secure initial passwords for new admin accounts and send them via email.

REQ-SUPERADMIN-4: System shall require new administrators to change their password on first login.

REQ-SUPERADMIN-5: System shall allow Super Admin to view all administrator accounts with their creation dates and last login times.

REQ-SUPERADMIN-6: System shall allow Super Admin to suspend or delete administrator accounts.

REQ-SUPERADMIN-7: System shall log all admin account management actions (create, suspend, delete) with Super Admin ID and timestamp.

System Configuration:

REQ-SUPERADMIN-8: System shall allow Super Admin to configure OTP expiry duration (default: 10 minutes).

REQ-SUPERADMIN-9: System shall allow Super Admin to configure session timeout durations per user type.

REQ-SUPERADMIN-10: System shall allow Super Admin to configure maximum file upload sizes per file type.

REQ-SUPERADMIN-11: System shall allow Super Admin to configure rate limiting thresholds for authentication endpoints.

REQ-SUPERADMIN-12: System shall allow Super Admin to configure automated diagnosis follow-up intervals.

REQ-SUPERADMIN-13: System shall log all configuration changes with before/after values.

REQ-SUPERADMIN-14: System shall validate all configuration changes before applying them.

Audit and Monitoring:

REQ-SUPERADMIN-15: System shall provide Super Admin with access to all system-wide audit logs.

REQ-SUPERADMIN-16: System shall allow filtering audit logs by user type, action type, date range, and user ID.

REQ-SUPERADMIN-17: System shall allow Super Admin to export audit logs in CSV or PDF format.

REQ-SUPERADMIN-18: System shall display system health metrics (active users, storage usage, API response times).

Security:

REQ-SUPERADMIN-19: System shall require Super Admin accounts to use MFA via email OTP for all logins.

REQ-SUPERADMIN-20: System shall enforce shortest session timeout for Super Admin accounts (default: 15 minutes).

REQ-SUPERADMIN-21: System shall prevent Super Admins from modifying or deleting audit logs.

REQ-SUPERADMIN-22: System shall prevent Super Admins from accessing or modifying patient medical records.

3.21 Administration Portal

3.21.1 Description and Priority

Description: The Administrator Portal provides regular admins with tools to verify and manage user registrations across all user types (patients, HCPs, labs, imaging centers). Admins review uploaded documents, validate identity information, and approve or reject registration requests. They can also view audit logs for monitoring purposes but cannot create other admin accounts or modify system-wide settings.

Priority: High

3.21.2 Stimulus/Response Sequences

Patient Verification:

1. Admin logs in → system displays Admin dashboard with pending verification counts
2. Admin navigates to "Pending Patient Verifications" → system displays list of unverified patient accounts with registration dates
3. Admin selects a patient account → system displays:
 - Submitted personal information (name, DOB, gender, national ID number, email, etc.)
 - Uploaded National ID images (front and back)
 - Uploaded selfie
4. Admin reviews documents and validates:
 - ID appears authentic
 - Selfie matches ID photo
 - Personal information matches ID
5. Admin either:
 - Approves: System marks patient as "Verified Patient" and sends confirmation email
 - Rejects: Admin provides rejection reason → system stores reason and notifies patient via email
6. System logs admin decision with admin ID, patient ID, action, reason (if rejected), and timestamp

HCP Verification:

7. Admin navigates to "Pending HCP Verifications" → system displays list of unverified HCP accounts
8. Admin selects an HCP account → system displays:
 - Personal information (name, DOB, specialization, license number, etc.)
 - Uploaded National ID
 - Uploaded medical license document
 - Uploaded profile photo (if provided)
9. Admin reviews documents and validates:
 - Medical license appears valid and matches license number
 - Name matches across all documents
 - License expiry date is valid
10. Admin approves or rejects with reason → system updates HCP status and sends notification
11. System logs admin decision

Lab/Imaging Center Verification:

12. Admin navigates to "Pending Lab/Imaging Verifications" → system displays list of unverified centers
13. Admin selects a center → system displays: - Center information (name, registration number, address, admin contact) - Uploaded accreditation/registration documents
14. Admin reviews documents and validates authenticity
15. Admin approves or rejects with reason → system updates center status and sends notification
16. System logs admin decision

Audit Log Viewing:

17. Admin navigates to "Audit Logs" → system displays logs with filtering options: - By user type - By action type (limited to verification actions and data access logs) - By date range
18. Admin can view but not export or modify logs

3.21.3 Functional Requirements

User Verification:

REQ-ADMIN-1: System shall display separate queues for pending patient, HCP, lab, and imaging center verifications.

REQ-ADMIN-2: System shall display all submitted information and uploaded documents for verification review.

REQ-ADMIN-3: System shall allow admins to approve or reject registrations.

REQ-ADMIN-4: System shall require a rejection reason when an admin rejects a registration.

REQ-ADMIN-5: System shall send email notifications to users upon approval or rejection of their registration.

REQ-ADMIN-6: System shall update user account status immediately upon admin decision (Verified or Rejected).

REQ-ADMIN-7: System shall log all verification decisions with admin ID, user ID, decision, reason, and timestamp.

REQ-ADMIN-8: System shall display verification statistics (total pending, approved today, rejected today) on the dashboard.

Document Review:

REQ-ADMIN-9: System shall display uploaded documents (National ID, medical licenses, accreditation documents) in a reviewable format.

REQ-ADMIN-10: System shall allow admins to zoom, rotate, and download uploaded documents for closer inspection.

REQ-ADMIN-11: System shall display submitted form data alongside uploaded documents for cross-validation.

Audit Log Access:

REQ-ADMIN-12: System shall allow admins to view audit logs filtered by user type, action type, and date range.

REQ-ADMIN-13: System shall restrict admin audit log access to verification actions and data access events only.

REQ-ADMIN-14: System shall not allow admins to export, modify, or delete audit logs.

Access Restrictions:

REQ-ADMIN-15: System shall prevent admins from creating other administrator accounts.

REQ-ADMIN-16: System shall prevent admins from modifying system-wide configuration settings.

REQ-ADMIN-17: System shall prevent admins from accessing or modifying patient medical records.

REQ-ADMIN-18: System shall prevent admins from accessing full patient medical histories (only registration info visible).

Authentication:

REQ-ADMIN-19: System shall require admin accounts to use email and password authentication.

REQ-ADMIN-20: System shall support optional MFA via email OTP for admin accounts.

REQ-ADMIN-21: System shall enforce a 30-minute session timeout for admin accounts.

REQ-ADMIN-22: System shall require admins to change their password on first login.

REQ-ADMIN-23: System shall log all admin authentication events (login, logout, failed attempts).

4. External Interface Requirements

4.1 User Interfaces

The Sijill system provides multiple user-facing interfaces tailored to the needs of different user classes. User interface design specifications, including detailed wireframes, mockups, and interaction patterns, will be documented separately in dedicated UI/UX design documentation created using Figma.

- **4.1.1 Patient Mobile Application**

Platform: Cross-platform mobile application developed using Flutter, supporting Android and iOS devices.

Key Interface Requirements:

- Bottom navigation pattern for primary sections: Dashboard, Medical Records, Notifications, and Profile
- QR code display interface for emergency medical access card
- Chronological timeline view for medical history browsing
- OTP generation interface for granting temporary provider access
- Push notification interface for medication and appointment reminders
- Secure authentication screens (login, registration, password recovery)

Interface Constraints:

- Must support common mobile screen sizes (minimum 5-inch displays)
- Must be usable in portrait orientation (landscape support optional)
- Critical actions (OTP generation, emergency access) must be accessible within 3 taps from the home screen

- **4.1.2 Healthcare Provider Web Portal**

Platform: React.js-based web application accessible via modern desktop browsers (Chrome, Firefox, Safari, Edge - latest two versions).

Key Interface Requirements:

- Top navigation bar with primary workflow access
- Patient search and selection interface
- Multi-step encounter documentation form with progressive navigation (symptoms → diagnosis → medications → orders → notes)
- Medical coding search interfaces for ICD-11 (diagnoses) and SNOMED CT (symptoms)
- Read-only patient medical history viewer
- OTP validation interface for accessing full patient records

Interface Constraints:

- Optimized for desktop/laptop screens (minimum 1366x768 resolution)
- Form validation with inline error messages
- Session timeout warnings before automatic logout
- **4.1.3 Laboratory Portal**

Platform: React.js-based web application.

Key Interface Requirements:

- Order Token and OTP validation interface
- Structured data entry forms for lab test results
- File upload interface supporting PDF reports and image attachments (JPEG, PNG)
- Upload progress indicators and validation feedback
- **4.1.4 Imaging Center Portal**

Platform: React.js-based web application.

Key Interface Requirements:

- Order Token and OTP validation interface
- DICOM and standard image file upload interface with drag-and-drop support
- Radiology report upload (PDF format)
- Upload progress indicators for large file handling
- **4.1.5 Research Department Portal**

Platform: React.js-based web application.

Key Interface Requirements:

- Dashboard with data visualizations (charts, graphs, statistics)
- Filtering and query builder interface for aggregated anonymized data
- Data export interface (CSV, Excel formats)
- **4.1.6 Administrative Dashboard**

Platform: React.js-based web application.

Key Interface Requirements:

- User management interface (approve/reject HCPs, labs, imaging centers)
- System configuration interface
- Audit log viewer with filtering and search capabilities
- Account verification queue with document review interface
- **4.1.7 Common Interface Requirements**

All web portals shall:

- Use HTTPS for all communications
- Display clear session status and timeout warnings
- Provide logout functionality accessible from all screens
- Show loading states during asynchronous operations
- Display error messages in a consistent, user-friendly manner
- Support keyboard navigation for accessibility

All interfaces shall:

- *Use Arabic and English language support (interface language can be configured per deployment)*
- *Display timestamps in the user's local timezone*
- *Follow consistent terminology as defined in the Glossary (Appendix A)*

4.2 Hardware Interfaces

The Sijill system does not require any specialized hardware interfaces. The system is a pure software solution that operates on standard commercial devices:

- *Mobile devices (smartphones) with camera capabilities for QR code scanning and document capture*
- *Desktop/laptop computers for web portal access*
- *Standard server infrastructure for backend services*

All hardware interactions occur through standard operating system APIs and do not require custom drivers or specialized hardware components.

4.3 Software Interfaces

4.3.1 Backend Runtime Environment

Node.js Runtime

- *Version: 20.x LTS or later*
- *Purpose: Execute backend application services*
- *Interface: Standard Node.js runtime APIs*

NestJS Framework

- *Version: 10.x or later*
- *Purpose: Backend application framework providing REST API endpoints, dependency injection, and service architecture*
- *Interface: HTTP/HTTPS RESTful APIs*

4.3.2 Database Management System

PostgreSQL

- *Version: 16.x LTS or later*
- *Purpose: Primary relational database for storing structured medical records, user accounts, authentication data, and system metadata*
- *Interface: Standard PostgreSQL wire protocol via database driver*
- *Data Shared: Patient records, clinical encounters, user profiles, access logs, orders, lab results, imaging metadata*
- *Communication: Connection pooling via application-level database client library*

4.3.3 File Storage

Local Filesystem

- *Purpose: Storage for uploaded files including:*
 - *PDF documents (lab reports, radiology reports, identity documents)*
 - *Medical images (JPEG, PNG)*
 - *DICOM imaging files*
 - *Patient and provider identity verification documents*
- *Interface: Standard POSIX filesystem APIs via Node.js fs module*
- *Security: Files stored with restricted permissions; access mediated through backend APIs only*
- *Organization: Hierarchical directory structure organized by entity type and UUID*

4.3.4 Mobile Application Framework

Flutter SDK

- *Version: 3.x stable channel or later*
- *Purpose: Cross-platform mobile application framework for patient-facing mobile app*
- *Target Platforms: Android 7.0+ (API level 24+), iOS 12.0+*
- *Interface: Platform channels for native device capabilities (camera, local storage, push notifications)*

4.3.5 Web Application Framework

React.js

- *Version: 18.x LTS or later*
- *Purpose: Frontend framework for all web-based portals (HCP, Lab, Imaging, Admin, Research)*
- *Interface: Communicates with backend via RESTful HTTP/HTTPS APIs*
- *Browser Requirements: Latest two major versions of Chrome, Firefox, Safari, and Edge*

4.3.6 External Medical Terminology Services

ICD-11 API (International Classification of Diseases)

- *Provider: World Health Organization (WHO)*
- *Endpoint: <https://icd.who.int/icdapi>*
- *Purpose: Search and validate diagnosis codes during clinical encounter documentation*
- *Communication: HTTPS REST API*
- *Data Exchange: Search queries (diagnosis keywords) → JSON responses (ICD-11 codes, descriptions)*
- *Authentication: API key-based (to be obtained during implementation)*

SNOMED CT Terminology Services

- *Provider: SNOMED International or compatible terminology server*
- *Purpose: Search and validate clinical terms for symptom documentation*
- *Communication: HTTPS REST API*
- *Data Exchange: Search queries (symptom keywords) → JSON responses (SNOMED CT codes, preferred terms)*
- *Authentication: API key-based or publicly accessible terminology browser*

4.3.7 Identity Verification Services

Optical Character Recognition (OCR)

- *Purpose: Extract structured data from uploaded identity documents (National ID, medical licenses)*
- *Implementation: To be determined during development phase; candidates include:*
 - *Open-source: Tesseract OCR*
 - *Cloud services: AWS Textract, Google Cloud Vision API, Azure Computer Vision*
- *Interface: Image input → structured text output (JSON format)*
- *Data Extracted: Names, ID numbers, dates, license information*

Face Matching

- *Purpose: Verify identity by comparing selfie photos against ID card photos during registration*
- *Implementation: To be determined during development phase; candidates include:*
 - *Open-source libraries: face-recognition, OpenCV*
 - *Cloud services: AWS Rekognition, Azure Face API*
- *Interface: Two image inputs → similarity score/match result*
- *Constraint: Must achieve acceptable accuracy for identity verification use case*

4.3.8 Communication Services

SMS/OTP Delivery

- *Purpose: Send one-time passwords for phone verification, multi-factor authentication, and password recovery*
 - *Implementation: To be determined during development phase; candidates include:*
 - *Third-party providers: Twilio, AWS SNS, MessageBird*
 - *Local telecom provider APIs (region-specific)*
 - *Interface: HTTPS REST API*
 - *Data Exchange: Phone number + message content → delivery confirmation*
- Push Notifications (Mobile)**
- *Purpose: Deliver medication reminders, appointment reminders, and access request notifications to patient mobile app*
 - *Service: Firebase Cloud Messaging (FCM) for Android and Apple Push Notification Service (APNs) for iOS*
 - *Interface: Backend sends notification payload via Firebase Admin SDK or APNS HTTP/2 API*
 - *Data Exchange: Device tokens, notification payloads → delivery receipts*
- Email Notifications**
- *Purpose: Account verification, password recovery, administrative notifications*
 - *Implementation: To be determined; candidates include SMTP relay, SendGrid, AWS SES*
 - *Interface: SMTP or HTTPS REST API*
 - *Data Exchange: Email address, subject, body → delivery status*

4.3.9 Security & Encryption

TLS/HTTPS

- *Version: TLS 1.2 or later (TLS 1.3 preferred)*
- *Purpose: Encrypt all client-server communications*
- *Scope: All HTTP API traffic, web portal access, mobile app backend communication*

Password Hashing

- *Library: bcrypt or Argon2*
- *Purpose: Securely hash and store user passwords*
- *Interface: Application-level library providing hash and compare functions*

4.3.10 DICOM Support

DICOM File Handling

- *Standard: DICOM 3.0*
- *Purpose: Accept, validate, and store medical imaging files from imaging centers*
- *Implementation: DICOM parsing library (e.g., dcm4che, dicom-parser, or cornerstone.js for web viewing)*
- *Interface: File upload → validation → metadata extraction → secure storage*
- *Constraint: Must validate DICOM compliance before accepting files*

4.3.11 Data Exchange Format

REST API Communication

- *Protocol: HTTP/HTTPS*
- *Format: JSON for structured data exchange*
- *Authentication: JWT (JSON Web Tokens) for session management*
- *Standard Headers: Content-Type, Authorization, Accept*

Inter-Component Data Sharing

- *All components communicate via the centralized backend REST API*
- *No direct database access from frontend applications*

- *Shared data (patient records, orders, results) accessed exclusively through authenticated API endpoints*
- *State management on clients handled in-memory (React state, Flutter state management)*

Notes:

- *Specific versions and service providers for external APIs (OCR, face matching, SMS) will be finalized during the detailed design and implementation phases based on availability, cost, and performance requirements.*
- *All external API integrations shall include error handling, retry logic, and fallback mechanisms.*
- *API documentation for internal REST endpoints will be maintained separately using OpenAPI/Swagger specification.*

4.4 Communications Interfaces

4.4.1 Network Protocols

HTTP/HTTPS

- *Protocol: HTTPS (HTTP over TLS/SSL) for all client-server communication*
- *Version: HTTP/1.1 minimum, HTTP/2 preferred for improved performance*
- *Port: Standard port 443 for HTTPS*
- *Purpose: All API requests, web portal access, mobile app backend communication*
- *Security: Mandatory TLS 1.2 or later; plain HTTP connections shall be rejected or redirected to HTTPS*

WebSocket (Optional)

- *Purpose: Real-time notifications and updates (if implemented for push notifications to web portals)*
- *Protocol: WSS (WebSocket Secure) over TLS*
- *Fallback: Long polling or server-sent events if WebSocket is unavailable*

4.4.2 API Communication

RESTful API Architecture

- *Style: REST (Representational State Transfer)*
- *Data Format: JSON (JavaScript Object Notation)*
- *Character Encoding: UTF-8*
- *Authentication: Bearer token authentication using JWT (JSON Web Tokens)*
- *Request Methods: GET, POST, PUT, PATCH, DELETE*
- *Status Codes: Standard HTTP status codes (200, 201, 400, 401, 403, 404, 500, etc.)*

4.4.3 File Upload Communication

Multipart Form Data

- *Content-Type: multipart/form-data*
- *Purpose: Upload PDF documents, medical images, DICOM files, identity documents*
- *Maximum File Size: Configurable per file type (e.g., 50MB for DICOM, 10MB for PDFs, 5MB for images)*
- *Supported Formats:*
 - *Images: JPEG, PNG*
 - *Documents: PDF*
 - *Medical Imaging: DICOM (.dcm)*

- *Transfer: Chunked transfer encoding for large files*
- *Validation: File type validation (MIME type and file signature), virus scanning before storage*

4.4.4 Email Communication

SMTP Protocol

- *Purpose: Send account verification emails, password recovery emails, administrative notifications*
- *Security: SMTP over TLS (STARTTLS) or SMTPS*
- *Port: 587 (STARTTLS) or 465 (SMTPS)*
- *Authentication: Username/password or API key authentication*
- *Message Format: MIME multipart (text/plain and text/html alternatives)*

Email Message Standards

- *Encoding: UTF-8*
- *Format: RFC 5322 compliant*
- *Content: Plain text with optional HTML formatting*
- *Sender Verification: SPF, DKIM, and DMARC records configured for domain authentication*

4.4.5 Push Notifications

Mobile Push Notifications

- *Android: Firebase Cloud Messaging (FCM)*
 - *Protocol: HTTPS*
 - *Format: JSON payload*
 - *Maximum payload size: 4KB*
- *iOS: Apple Push Notification Service (APNs)*
 - *Protocol: HTTP/2-based API*
 - *Format: JSON payload*
 - *Maximum payload size: 4KB*

4.4.6 External API Integration

Medical Terminology APIs (ICD-11, SNOMED CT)

- *Protocol: HTTPS REST*
- *Authentication: API key in request headers or OAuth 2.0*
- *Request Format: JSON or URL query parameters*
- *Response Format: JSON*
- *Rate Limiting: Respect rate limits imposed by external providers*
- *Caching: Cache frequently accessed codes locally to reduce external API calls*
- *Timeout: 30-second request timeout with retry logic*

OCR and Face Matching Services

- *Protocol: HTTPS REST*
- *Authentication: API key or OAuth 2.0*
- *Request Format: Base64-encoded image data in JSON or multipart/form-data*
- *Response Format: JSON with extracted data or match confidence scores*
- *Timeout: 60-second timeout for processing-intensive operations*

4.4.7 Security Requirements

Encryption in Transit

- *All network communications shall use TLS 1.2 or higher*
- *Certificate validation shall be enforced on all clients*
- *Self-signed certificates shall not be accepted in production environments*
- *Perfect Forward Secrecy (PFS) cipher suites shall be preferred*

Authentication Tokens

- *JWT tokens shall include expiration time (configurable, default 1 hour for patients, 30 minutes for providers/labs/imaging centers)*
- *Refresh tokens shall be used for session extension without re-authentication*
- *Tokens shall be transmitted only in Authorization headers, never in URLs*

Data Sanitization

- *All input data shall be validated and sanitized before processing*
- *Output encoding shall be applied to prevent injection attacks*
- *File uploads shall be scanned for malware before storage*

4.4.8 Network Requirements**Bandwidth**

- *Minimum: 1 Mbps for basic portal operations*
- *Recommended: 5+ Mbps for DICOM file uploads and medical image viewing*

Latency

- *Target API response time: < 500ms for typical requests*
- *Acceptable: < 2 seconds for database-intensive operations*
- *File uploads: Progress indicators for operations exceeding 3 seconds*

Availability

- *Backend services shall be designed for 99.5% uptime*
- *Graceful degradation: Client applications shall handle temporary network failures with appropriate retry mechanisms and user feedback*

Firewall and Network Configuration

- *Outbound HTTPS (port 443) required for all external API calls*
- *Inbound HTTPS (port 443) required for backend server*
- *No requirement for specialized ports or protocols beyond standard HTTPS*

4.4.9 Synchronization and Concurrency**Data Synchronization**

- *Mobile app shall sync with backend on app launch and after network reconnection*
- *Real-time updates not required; polling acceptable for non-critical updates*
- *Optimistic UI updates with background sync and conflict resolution*

Concurrent Access Handling

- *Last-write-wins for profile updates (with timestamp tracking)*
- *Append-only for medical records and audit logs (no conflicts)*
- *Database-level locking for critical operations (e.g., OTP validation, order fulfillment)*

5. Other Nonfunctional Requirements

5.1 Performance Requirements

API Response Times

- *Standard API requests (authentication, data retrieval): < 2 seconds under normal load*
- *Complex queries (medical history retrieval, filtered searches): < 5 seconds*
- *File uploads: Progress feedback required for operations exceeding 3 seconds*

Concurrency

- *System shall support minimum 100 concurrent users across all portals*
- *Target capacity: 500 concurrent users for production deployment*

File Handling

- *DICOM file uploads up to 50MB with chunked transfer*
- *PDF reports up to 10MB*

- Standard images up to 5MB

Database Performance

- Patient record queries shall return results within 1 second for typical medical history sizes (< 1000 encounters)

5.2 Safety Requirements

Data Integrity

- System shall prevent accidental deletion or modification of medical records through immutable storage mechanisms
- All critical operations (prescription creation, diagnosis documentation, result uploads) shall require explicit confirmation
- System shall maintain audit logs to trace all data access and modifications for accountability

Emergency Access

- Emergency medical card access shall remain functional even during partial system outages
- Critical patient information (allergies, blood type, chronic conditions) shall be retrievable within 5 seconds via QR code scan

Error Prevention

- System shall validate all medical codes (ICD-11, SNOMED CT) before saving
- System shall prevent submission of incomplete encounter records through mandatory field validation
- System shall display clear warnings before irreversible actions

5.3 Security Requirements

Security requirements are extensively specified throughout Section 3 (functional requirements) and include:

- Multi-factor authentication via OTP (REQ-AUTH-7, REQ-HCP-AUTH-8)
- Encrypted data transmission via HTTPS/TLS (REQ-REG-11, Section 4.4.8)
- Role-based access control with patient-granted temporary access (REQ-HCP-MRM-1)
- Immutable audit logging (REQ-LOG-3, REQ-LOG-7)
- Secure password storage using hashing (REQ-REG-11, REQ-HCP-REG-11)
- Rate limiting and account lockout protection (REQ-AUTH-12, REQ-HCP-AUTH-11)

Additional Security Constraints:

- All uploaded files shall undergo malware scanning before storage
- Session tokens shall expire after configurable timeout (default: 1 hour for patients, 30 minutes for providers)
- System shall enforce HTTPS; plain HTTP connections shall be rejected

5.4 Software Quality Attributes

Reliability

- Target system availability: 99.5% uptime during operational hours
- Mean Time Between Failures (MTBF): > 720 hours
- Graceful degradation: System shall handle database connection failures with appropriate error messages and retry mechanisms

Usability

- Patient-facing mobile app shall require no more than 3 taps to access critical features (emergency card, OTP generation)

- Healthcare provider encounter documentation shall follow clinical workflow patterns to minimize data entry time
- Error messages shall be clear, specific, and actionable

Maintainability

- Code shall follow consistent style guidelines and include inline documentation
- Database schema shall use clear naming conventions and include migration scripts
- API endpoints shall be documented using OpenAPI/Swagger specification

Testability

- All functional requirements shall be verifiable through automated or manual test cases
- System shall provide test environments separated from production data
- Authentication and authorization logic shall be unit-testable

Portability

- Backend services shall run on standard Linux environments (Ubuntu 20.04+)
- Mobile application shall support Android 7.0+ and iOS 12.0+
- Web portals shall function on latest two major versions of modern browsers

Interoperability

- System shall use standard medical coding systems (ICD-11, SNOMED CT) to facilitate future integration
- DICOM file handling shall comply with DICOM 3.0 standard
- API design shall follow RESTful principles for potential third-party integration

6. Other Requirements

6.1 Implementation Context and Constrains

This SRS describes a graduation project implementation of the Sijill system. Several design decisions reflect the academic and prototype nature of this project:

Identity Verification Approach *In a production deployment operated by a governmental health authority (e.g., Ministry of Health), the system would integrate with official government identity verification APIs and medical licensing databases to perform automated, real-time verification of:*

- Patient national ID validation against civil registry databases
- Healthcare provider medical license verification against official medical council databases
- Laboratory and imaging center accreditation verification against health authority registries

Due to the unavailability of such APIs in the academic project context, this implementation uses a hybrid approach combining automated document processing (OCR, face matching) with manual approval workflows performed by system administrators. This manual approval step simulates the role that would be fulfilled by official API integrations in a real-world deployment.

Deployment Scope *This implementation targets a university medical administration pilot deployment with a limited user base. A nationwide production system would require additional considerations including:*

- High-availability infrastructure with geographic redundancy
- Compliance with national health data protection regulations
- Integration with existing national health information exchange systems
- Scalability to support millions of users and healthcare facilities

6.2 Legal and Compliance Considerations

The Sijill system, as conceptualized in this document, is designed explicitly as **national critical infrastructure for electronic health records**. Its implementation, operation, and data stewardship are intended to fall exclusively under the authority and ownership of a governmental body, such as the Ministry of Health and Population.

1. Governmental Ownership and Stewardship

- The Sijill platform, all associated software, and the aggregated health data within it shall be the property of the state.
- The government, acting through a designated authority, shall be the sole data controller and steward. This centralizes accountability and ensures that data is managed in the public interest, not for commercial gain.

2. Alignment with National Digital Health Strategy

- This project is conceived as a foundational pillar for Egypt's digital transformation in healthcare. It aims to eliminate fragmented health information, reduce medical errors, and provide a data-driven foundation for public health policy and epidemiological research.
- By creating a lifelong, portable health record for every citizen, Sijill directly supports national goals of improving healthcare quality, accessibility, and efficiency.

This Software Requirements Specification (SRS) is a purely academic deliverable. It represents a conceptual design and technical proposal for a graduation project at the Faculty of Computer Science & Artificial Intelligence. The authors, contributors, and affiliated academic institutions explicitly state and warrant the following:

1. **No Implementation Intent:** The project team, "OverEngineered," has **no intention, plan, or capability** to develop, deploy, or operate the Sijill system in any live, production, or public environment. This project is strictly confined to an academic demonstration and proof-of-concept within a controlled, non-production setting.
2. **Conditional on Governmental Authorization:** Any potential future realization of a system based on this concept is **expressly and unconditionally contingent** upon direct initiation, full funding, explicit written authorization, and complete operational control by the competent Egyptian governmental authorities, specifically the **Ministry of Health and Population**. Without such formal state sanction, this concept will not proceed beyond its academic prototype phase.
3. **No Data Responsibility:** The team assumes **zero liability or responsibility** for the collection, storage, or processing of any real citizen medical data. Any testing during the academic project will use fully anonymized, synthetic, or mock data that does not pertain to real individuals.
4. **Conceptual Nature:** This document is a theoretical blueprint. The specifications within are subject to change and would require rigorous validation, security auditing, and legal review by appropriate government bodies before any consideration for real-world use.
5. **Intellectual Property for Public Good:** The intellectual property embodied in this SRS is presented as a contribution to the public discourse on improving Egypt's healthcare infrastructure. It is offered to the relevant national authorities for their consideration, without any commercial claims or expectations from the authors.

In summary, this document should in no way be construed as a plan for an independent software product or service. It is an academic exercise outlining a potential solution that, by its very nature, can only be legitimately realized by and under the sovereign control of the Egyptian state.

7. Appendix A: Glossary

Acronyms and Abbreviations

Term	Definition
API	Application Programming Interface
DICOM	Digital Imaging and Communications in Medicine - Standard for medical imaging files
EHR	Electronic Health Record
HCP	Healthcare Provider (Doctor, Clinician)
HTTPS	Hypertext Transfer Protocol Secure
ICD-11	International Classification of Diseases, 11th Revision - WHO standard for disease coding
JWT	JSON Web Token - Authentication token standard
LOINC	Logical Observation Identifiers Names and Codes - Lab test coding standard
MFA	Multi-Factor Authentication
OCR	Optical Character Recognition
OTP	One-Time Password - Temporary access code
PDF	Portable Document Format
REST	Representational State Transfer - API architectural style
SNOMED CT	Systematized Nomenclature of Medicine Clinical Terms - Clinical terminology standard
SMS	Short Message Service
SRS	Software Requirements Specification
TLS	Transport Layer Security
UUID	Universally Unique Identifier
WHO	World Health Organization

Domain-Specific Terms

Term	Definition
Active Diagnosis	A medical diagnosis that remains unresolved and is tracked for outcome monitoring
Clinical Encounter	A documented interaction between a healthcare provider and patient, including symptoms, diagnosis, and treatment
Emergency Access Card	QR code-based mechanism providing immediate access to critical patient information without authentication
Immutable Log	Audit record that cannot be modified or deleted after creation, ensuring accountability
Medical Record	Comprehensive collection of a patient's health information including visits, diagnoses, prescriptions, lab results, and imaging studies
Order Token	Unique identifier for a lab or imaging test order created by a healthcare provider
Patient Portal	Mobile application interface allowing patients to access their medical records and manage access permissions
Provider Portal	Web interface allowing healthcare providers to document clinical encounters and access patient records
Self-Requested Diagnostic	Lab or imaging test initiated directly by a patient without a doctor's order
Upload Session	Time-limited, scope-restricted session allowing labs or imaging centers to submit results for a specific order

Verified User	User account (patient, provider, lab, imaging center) that has completed identity verification and is authorized to access system features
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Workflow-Specific Terms

Term	Definition
Doctor-Initiated Order	Lab or imaging test ordered by a healthcare provider during a clinical encounter
OTP-Granted Access	Temporary access to patient records granted via one-time password approval by the patient
Patient-Generated Data	Health notes and observations entered by patients themselves, distinct from clinical records
Single-Use Token	Authentication token (OTP or Order Token) that becomes invalid after one successful use
Upload-Only Access	Restricted permission level allowing only submission of results without viewing patient history